

		$\approx 300 \text{ m/s}$
17	A	Let the speed of the wave be v. The wavelength $\lambda = \frac{v}{f} = \frac{v}{12.5} \rightarrow v = 12.5\lambda$ For the displacement to be zero at Q, the wave travels a distance $d = \frac{\lambda}{8} = 0.125\lambda$. The time taken for this $= \frac{d}{v} = \frac{0.125\lambda}{12.5\lambda} = 0.010 \text{ s}$

Qn	Ans	Solution
18	D	At lowest frequency